

You are more than welcome to openly discuss these problems. You don't need to hand in these problems. The home assignments are graded only through quizzes. Questions with [For Fun] mark will not enter the quizzes.

Home assignment 1

Enters in the quizzes during the week 3: 26 January.

1. The semi-annual y_t is modelled by $ETS(AAA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 4) \\ s_t = s_{t-2} + 0.1u_t \\ b_t = b_{t-1} + 0.2u_t \\ \ell_t = \ell_{t-1} + b_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + b_{t-1} + s_{t-2} + u_t \end{cases}$$

- (a) Given that $s_{100} = 2$, $s_{99} = -1.9$, $b_{100} = 0.5$, $\ell_{100} = 4$ find 95% predictive interval for y_{102} .
 (b) In this problem particular values of parameters are specified. How many parameters are estimated in semi-annual $ETS(AAA)$ model before real forecasting?

2. The $ETS(AAdN)$ model is given by the system

$$\begin{cases} u_t \sim \mathcal{N}(0; 16) \\ b_t = 0.9b_{t-1} + 0.2u_t \\ \ell_t = \ell_{t-1} + 0.9b_{t-1} + 0.1u_t \\ y_t = \ell_{t-1} + 0.9b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 20$ and $b_{100} = 2$.

- (a) Find the 95% predictive interval for y_{101} .
 (b) Find conditional probability $\mathbb{P}(y_{102} > 30 \mid \ell_{100}, b_{100})$.
 (c) Approximately find the best point forecast for y_{10000} .
 (d) Find the 95% predictive interval for b_{10000} .

3. The semi-annual y_t is modelled by $ETS(AAA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 4) \\ s_t = s_{t-2} + 0.1u_t \\ b_t = b_{t-1} + 0.2u_t \\ \ell_t = \ell_{t-1} + b_{t-1} + 0.3u_t \\ y_t = \ell_{t-1} + b_{t-1} + s_{t-2} + u_t \end{cases}$$

Given that $s_0 = 2$, $s_{-1} = -2$, $b_0 = 0.5$, $\ell_0 = 4$ decompose $y_1 = 3$, $y_2 = 6$, $y_3 = 4$ into trend, seasonal component and random shocks.

Home assignment 2

Enters in the quizzes during the week 5: 9 February.

1. Consider an $MA(2)$ process defined by the equation:

$$y_t = 5 + u_t + 2u_{t-1} + 4u_{t-2},$$

where (u_t) is a white noise process with variance $\text{Var}(u_t) = \sigma^2$.

- (a) Find the expected value $\mathbb{E}(y_t)$.
 - (b) Find the autocorrelation function $\rho(k) = \text{Corr}(y_t, y_{t-k})$.
 - (c) Is the process (y_t) stationary?
 - (d) Construct a 95% confidence interval for y_{101} , given $u_{100} = 1$ and $u_{99} = -1$.
2. Consider an $MA(1)$ process defined by the equation:

$$y_t = \mu + u_t + bu_{t-1},$$

where (u_t) is a white noise process with zero mean and variance σ^2 .

Assume we observe the following three consecutive realizations of this process:

$$y_1 = 2, \quad y_2 = 0.5, \quad y_3 = 1.5.$$

- (a) Using the method of moments, set up the system of equations by equating $\mathbb{E}(y_t)$, $\mathbb{E}(y_t^2)$ and $\mathbb{E}(y_t y_{t-1})$ with their sample counterparts. Derive estimates for the parameters μ , b and σ^2 .
 - (b) Based on your estimates, is the process invertible? Justify your answer.
3. Suppose it is known that the autocorrelation function of an MA process takes the following values:

$$\rho(1) = -\frac{13}{19}, \quad \rho(2) = -\frac{91}{190}, \quad \rho(k) = 0 \text{ for } k \geq 3.$$

- (a) What is the order of this MA process?
 - (b) Is there enough information to recover all parameters of the model generating this process?
 - (c) Recover the parameters if possible.
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Home assignment 3

Live in the quizzes: 2026-03-02.

1. Consider $MA(1)$ process $y_t = 5 + u_t + 2u_{t-1}$ where (u_t) is a white noise process.
 - (a) Find $\mathbb{E}(y_t)$.
 - (b) Find the autocorrelation function ρ_k .
 - (c) Find the first two values of the partial autocorrelation function, ϕ_{11}, ϕ_{22} .
 2. Consider $MA(2)$ process $y_t = 5 + u_t + 2u_{t-1} + 3u_{t-2}$ where (u_t) is a white noise process.
 - (a) Find $\mathbb{E}(y_t)$.
 - (b) Find the autocorrelation function ρ_k .
 - (c) Find the first two values of the partial autocorrelation function, ϕ_{11}, ϕ_{22} .
 3. Consider the stationary $AR(1)$ process with equation $y_t = 5 + 0.3y_{t-1} + u_t$ where (u_t) is a white noise process.
 - (a) Find $\mathbb{E}(y_t)$.
 - (b) Find the autocorrelation function ρ_k .
 - (c) Find the first two values of the partial autocorrelation function, ϕ_{11}, ϕ_{22} .
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Home assignment 4

Live in the quizzes: 2026-03-23.

1. Variables $y_1, y_2, \dots, y_n$ are independent and identically distributed with $\mathbb{E}(y_i^3) = \sqrt{a-7}$, where a is an unknown parameter.

Estimate a using the method of moments.

2. Variables $y_1, y_2, \dots, y_n$ are independent and identically distributed with density $f(y) = (a + 0.5)y^{a-0.5}$ for $y \in [0; 1]$ where a is an unknown parameter.

Estimate a using maximum likelihood.
